B.Sc. 4th Semester Zoology

Selected Practical Work

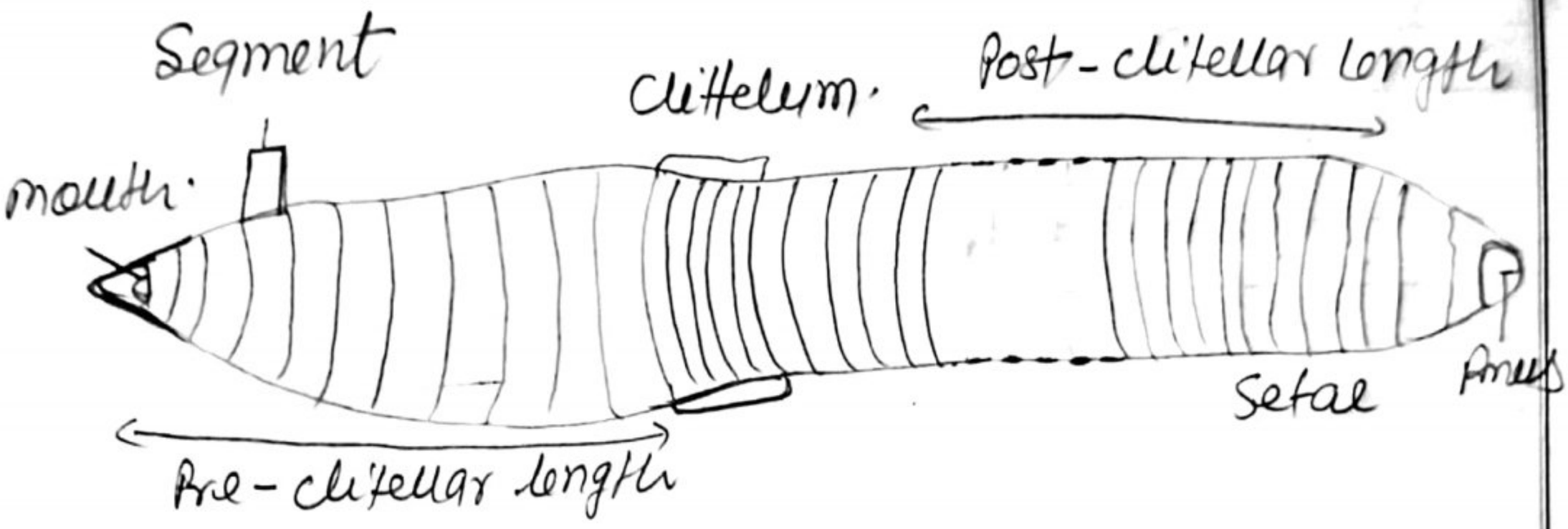
Shafaque Naushad

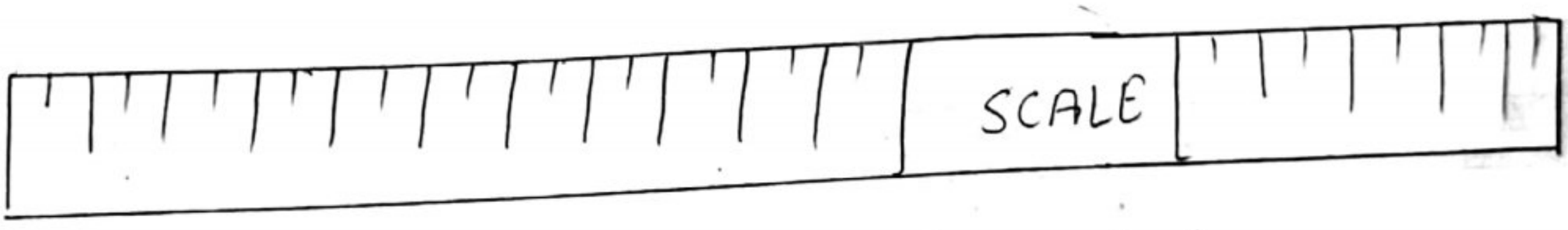
B.Sc. Life Science (Botany and Zoology) | Maharaja Suhel Dev University, Azamgarh

CONTENTS

Expt.	Experiment	What it covers
1	Earthworm pre- and post-clitellar length	Measurements of 5 earthworms; mean (3.68 and 7.56 cm), median, mode and standard deviation (1.16 and 1.82 cm).
6	Sequence analysis using NCBI BLAST	Downloading a gene sequence from NCBI and reading BLAST results: query details, alignment, similarity.
7	Phylogenetic analysis using PHYLIP	Basic terms and the workflow from a multiple sequence alignment to a distance matrix with Dnadist.
8	PCR primer design	Primer design from an NCBI gene sequence; Primer3 output with four candidate primer pairs.

Scanned pages from my handwritten semester 4 Zoology practical file. The other experiments in the file (ABO blood grouping, restriction enzyme digestion, PCR mutation detection and basic computer applications) are not included here.





Data collection setup

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Aim- To Measure the pre and post clitellar lengths of earthworms and calculate mean, median, mode standard deviation.

Procedure -: * Collect a few earthworms (n=5) from your local vicinity and bring them to the lab. you may keep them alive and release back to soil after experiment is over

* Carry one large scale with you and place the earthworm over the scale. Straiten it over the scale.

* Now, identify the clitellum as shown and measure the length from head to clitellum this is pre-clitellar length.

* Repeat the same process for measurment of length from clitellum to the end of tail. this is post clitellar length.

* Repeat the experiment for all the earthworm that your or your classmates have bought

Earthworm No	Pre clitellar length (cm)	Post clitellar length (cm)
1	5.2	8.8
2	4.5	8.4
3	3.2	7.3
4	3.2	8.8
5	2.3	4.5

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Calculation of mean - * Add all the values of pre-clitellar lengths and devided by number of samples (5)

$$(5.2 + 4.5 + 3.2 + 3.2 + 2.3) = 3.68$$

mean pre-clitellar length = 3.68 cm.

add all the values of pre-clitellar length and devide by number of sample (5)

$$\frac{8.8 + 8.4 + 7.3 + 8.8 + 4.5)}{5} = 7.56$$

mean post-clitellar length = 7.56 cm. Ans.

Calculation of median :- In order to calculate ever arrange the data in increasing order and then find the middle value so. if we have 5 samples in total. we will have 3rd values as middle value

Hence for pre-clitellar lengths (in increasing order)

2.3, 3.2, 3.2, 4.5, 5.2.

The middle value is 3rd or 3.2. hence it is the median.

similarly, for post clitellar lengths.

4.5, 7.3, 8.4, 8.8, 8.8.

The middle value is 3rd or 8.4. hence it is the median.

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- **Calculation of mode-** mode is the most frequent value of a sample
- In case of pre-clitellar length it is 3.2. which appeared two times.
In case of post-clitellar length it is 8.8 which appeared two times.

while all the values appeared only one time. it is possible that some data sets may have two different modes or even no mode in some cases.

- **Calculation of standard deviation-**

Prepare a table as described below. where Calculate (x-mean) for each value and then square all values and add them as shown below.

* Standard deviation for pre-clitellar lengths(x)

Earthworm no	x	x - mean	$(x-mean)^2$
1	5.2	5.2-3.68 =1.52	2.3104
2	4.5	4.5-3.68 = 0.82.	0.6724
3	3.2	3.2-3.68 = -0.48	0.2304
4	3.2	3.2-3.68 = -0.48	0.2304
5	2.3	2.3-3.68 = -1.38	1.9044
Total			5.348

Standard deviation = $\sqrt{\frac{5.348}{n-1}}$, where n is the number of samples (5 in the case)

So SD = $\sqrt{\frac{5.348}{4}}$ = $\sqrt{1.337}$ = 1.1563 cm.

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Standard deviation for past - cliffellar lengths (x)

Earth worms no	x	x - mean	$(x - mean)^2$
1	8.8	8.8 - 7.56 = 1.24	1.5376
2.	8.4	8.4 - 7.56 = 0.84	0.7056
3.	7.3	7.3 - 7.56 = 0.26	0.0676
4	8.8	8.8 - 7.56 = 1.24	1.5376
5	4.5	4.5 - 7.56 = -3.06	9.3636
Total			13.212

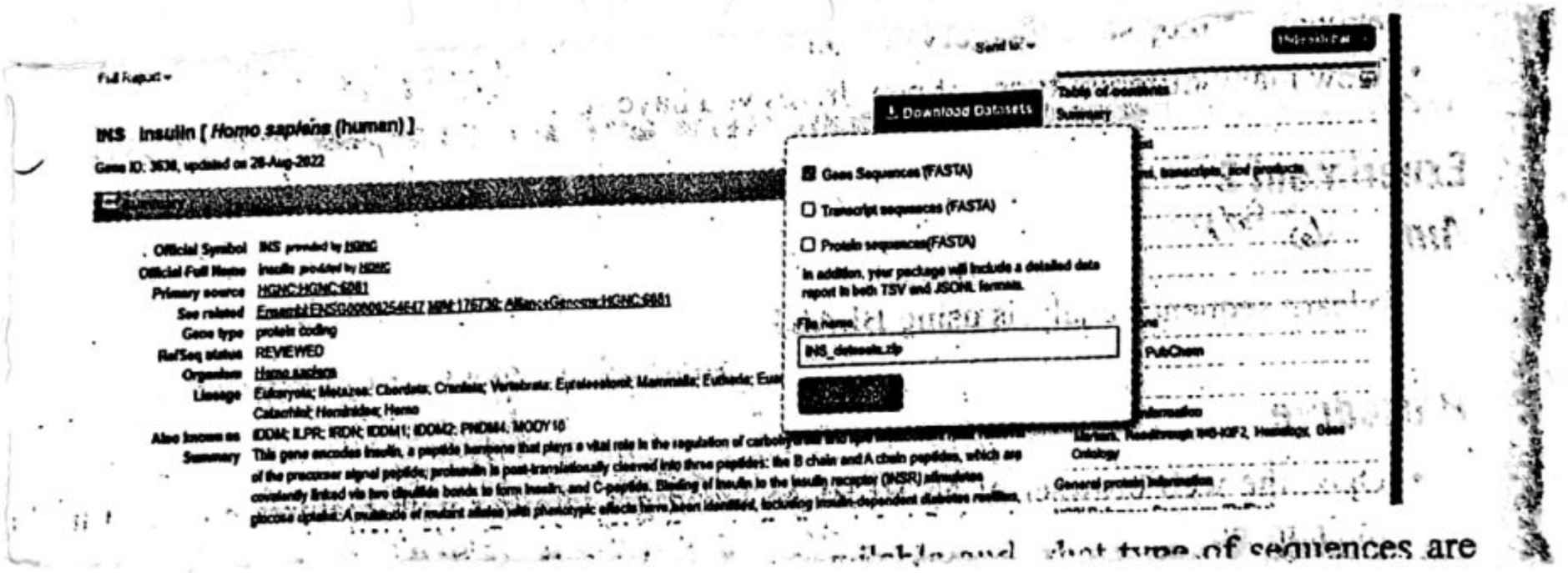
Standard deviation = $\sqrt{\frac{13.212}{n-1}}$, where n is the number of samples (5 in this case)

So SD = $\sqrt{\frac{13.212}{4}}$ = $\sqrt{3.303}$ = 1.8174 cm.

Ans

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...ckground what type of sequences are

Progress of the BLAST Process.

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Expt. No.....6..... Page No..... Date...../...../.....

Aim :- To learn sequence analysis using BLAST.

Procedure:-

* open the web browser of NCBI
* BLAST and get familiar with feature provided in the mainmenu.
* Learn basic procedure from the home page of BLAST from NCBI as shown below.
* Download a protein or D.N.A. sequence from NCBI gene bank, or randomly type a sequence by yourself.
* Now select a suitable tab nucleotide blast for DNA

Observation:-

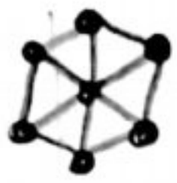
After submitting the query sequence for sequence similarity search, the result page will appear along with the information like Query id, Description, molecule type, Length of Sequne. Database name and BLAST program. Finally it shows the actual alignment, along with the query sequence on the top and database sequence below the query.

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testdata



Clique



Consense



Contml



Contrast



dnacomp



dnadist



dolpenny



drawgram-dll



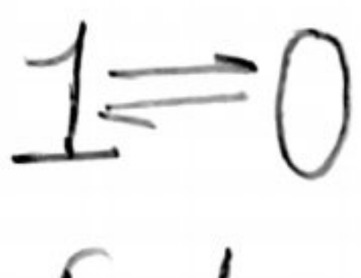
DrawGram



drawtree.dll



DrawTree



factor



Fitch

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Expt. No.....7..... Page No..... Date...../...../.....

Aim:- To learn about phylogenetic analysis using the programme PHYLIP.

Basic Terms to learn:-

1- Phylogenetic Analysis:- Analyze the evolutionary relationship betweens different organisms and this analysis would help to find out the change that occured in organisms during the evolution.

2- Tree topology:- Tree topology refers to the arrengment of phylogenetic tree.

Activity:-

1. To prepare phylogenetic tree, we need a multiple sequence aligened file, which is generally an output file form allietal software.
2. After the alignment process if completete, save the output file of multiple sequence alignment it in PHYLIP format as infile.phy.
3. Download the PHYLIP Program and use the sub programe called Dnadist in PHYLIP by clicking the icon and giving aforementioned infied as input.
4. After giving the correct input, if needed it will ask to change and settings for the programs by typing the first letter or number

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KEYS (in order of precedence):
>>>>>> left primer
<<<<<< right primer

ADDITIONAL OLIGOS

	start	len	tm	gc%	any_th	3'_th	hairpin	seq
1 LEFT PRIMER	474	20	58.96	50.00	0.00	0.00	0.00	TGCTTCGAGAACAGGTGAGT
RIGHT PRIMER	635	20	59.05	55.00	0.00	0.00	0.00	CAACAGGACCAACAGAGCAC
PRODUCT SIZE: 162, PAIR ANY_TH COMPL: 0.00, PAIR 3'_TH COMPL: 0.00								
2 LEFT PRIMER	454	20	58.95	55.00	0.00	0.00	0.00	TATCCACCATTCCTCGCCTC
RIGHT PRIMER	666	20	58.94	50.00	0.00	0.00	0.00	TGCCTGGGTTAGTGCTTACA
PRODUCT SIZE: 213, PAIR ANY_TH COMPL: 0.00, PAIR 3'_TH COMPL: 0.00								
3 LEFT PRIMER	296	20	58.75	50.00	0.00	0.00	0.00	GTGGCATGTGTGATTGGACA
RIGHT PRIMER	542	20	59.04	50.00	0.00	0.00	0.00	TACACAAGCTACACTCCGCA
PRODUCT SIZE: 247, PAIR ANY_TH COMPL: 16.94, PAIR 3'_TH COMPL: 11.20								
4 LEFT PRIMER	3232	20	58.75	50.00	0.00	0.00	0.00	GAGGTGGCTGACTTTGCATT
RIGHT PRIMER	3412	20	59.09	50.00	8.77	0.00	0.00	AACGGAGAGCATTAAGGCCT
PRODUCT SIZE: 181, PAIR ANY_TH COMPL: 2.58, PAIR 3'_TH COMPL: 0.00								

Statistics

	con sidered	too many Ns	in tar get	in excl reg	not ok reg	bad GC%	no GC clamp	tm too low	tm too high	high any_th compl	high 3'_th compl	high hair- pin	poly X	high end stab	lower case end	ok
Left	22059	0	0	0	0	5921	0	8275	1145	0	0	19	204	0	3510	2985
Right	22037	0	0	0	0	6194	0	8100	1127	0	0	8	211	0	3543	2854

Pair Stats:
considered 1067, unacceptable product size 1059, primer in pair overlaps a primer in a better pair 145, ok 8
libprimer3 release 2.4.0

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Expt. No. 8 Page No. Date ...../...../.....

Aim:- To learn how to perform primer designing for PCR.

Procedure:- 1. Download a gene sequence from NCBI - nuleotide, as we did for BLAST and MSA.
Some parameters which can be used for customization, such as if you want primers targeted at specific position in sequence, or you wish to exclude certain region
3. Once setup with all parameters, you can click on Pick primer button and wait for the primer suggestion.
4. You may finally look at the properties of primers as described in output and choose the suitable one.
5. Bingo! You have designed yours primer, now you can order them from a company and check them in PCR by yourself if they work fine.

Observation:-

After the succesful completion of parameter setup, the program provides an output of primer sets from the data as follows. Note that there can be more than the suggested primers and users have of further evalute them on their experimental choice.

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